

Alzheimer's Disease Detection

Using Deep Learning

Domain: Healthcare / Medical Image Classification

Dataset: OASIS Alzheimer's MRI (86,437 images)

Best Model: MobileNetV2 (Fine-Tuned) - 99.47% Accuracy

Platform: Google Colab (T4 GPU) + Flask Web Application

Date: April 2026

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1. Project Overview

This project is an end-to-end Alzheimer's disease detection system that uses deep learning models trained on brain MRI scans to classify the severity of Alzheimer's disease into four stages. The system includes a full machine learning pipeline for model training on Google Colab (T4 GPU) and a professional Flask web application for real-time prediction with clinical context.

Project Title	Alzheimer's Disease Detection System
Domain	Healthcare / Medical Image Classification
Technique	Deep Learning (CNNs & Transfer Learning)
Language	Python 3.12
Framework	TensorFlow 2.19 / Keras
Web App	Flask + SQLite + Bootstrap 5
Training	Google Colab with NVIDIA T4 GPU
Best Accuracy	99.47% (MobileNetV2 Fine-Tuned)

2. Problem Statement & Purpose

What is the Project?

This project develops an AI-powered system that can analyze brain MRI (Magnetic Resonance Imaging) scans and classify them into one of four stages of Alzheimer's disease progression: Non Demented, Very Mild Demented, Mild Demented, and Moderate Demented.

Why am I Doing It?

- Alzheimer's disease affects over 55 million people worldwide. Early and accurate detection is critical.
- Manual diagnosis by radiologists is time-intensive, subjective, and prone error based on the observer.
- AI-assisted detection can serve as a second opinion tool, improving diagnostic accuracy.
- The distinction between early stages is extremely subtle on MRI, making it ideal for deep learning.

What is the Purpose?

- Build a multi-class image classifier across four Alzheimer's stages.
- Compare 6 different deep learning architectures (CNN, MLP, VGG16, ResNet50, EfficientNetB0, MobileNetV2).
- Deploy the best model in a user-friendly web application with clinical context.

How Did I Do It?

- Collected the OASIS brain MRI dataset (86,437 images across 4 classes).
- Cleaned the data - validated all images, removed corrupted/duplicate files.

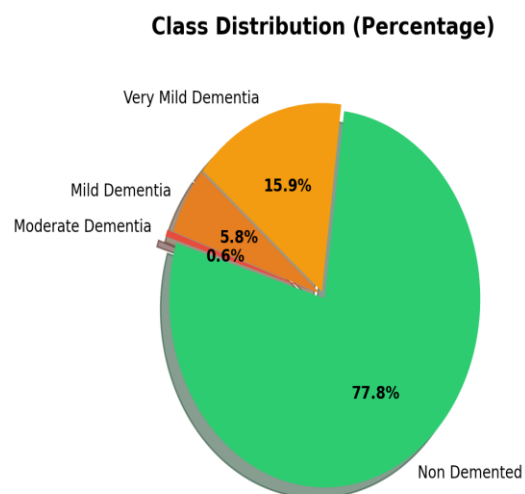
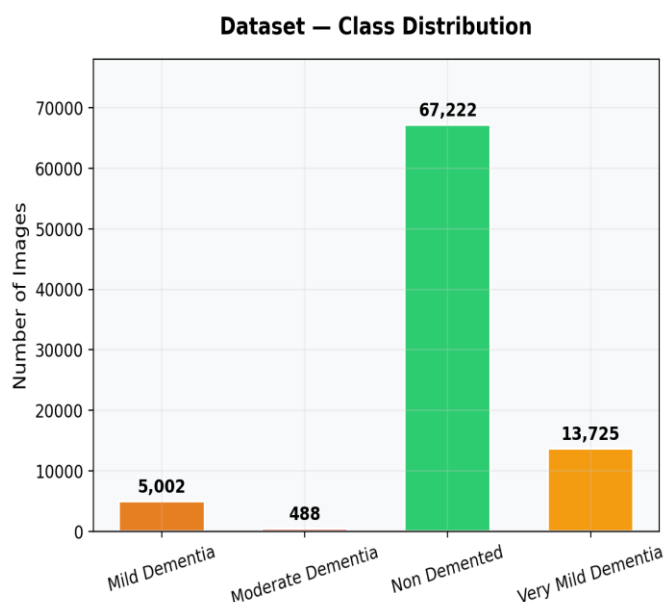
- Split the data into Train (70%), Validation (15%), Test (15%) with stratification.
- Applied data augmentation and computed class weights for imbalance handling.
- Trained 6 different models on Google Colab with a T4 GPU.
- Identified MobileNetV2 as the best base architecture.
- Fine-tuned MobileNetV2 with 224x224 input and partial backbone unfreezing - achieved 99.47% accuracy.
- Built a Flask web application with user auth, upload, prediction, and clinical results.

3. Dataset Description

The dataset used is the OASIS (Open Access Series of Imaging Studies) Alzheimer's Detection Dataset, sourced from Kaggle (ninadaithal/imagesoasis). It contains 86,437 labeled brain MRI scan images distributed across 4 classes. The dataset exhibits severe class imbalance, with 'Non Demented' having 138x more samples than 'Moderate Dementia'.

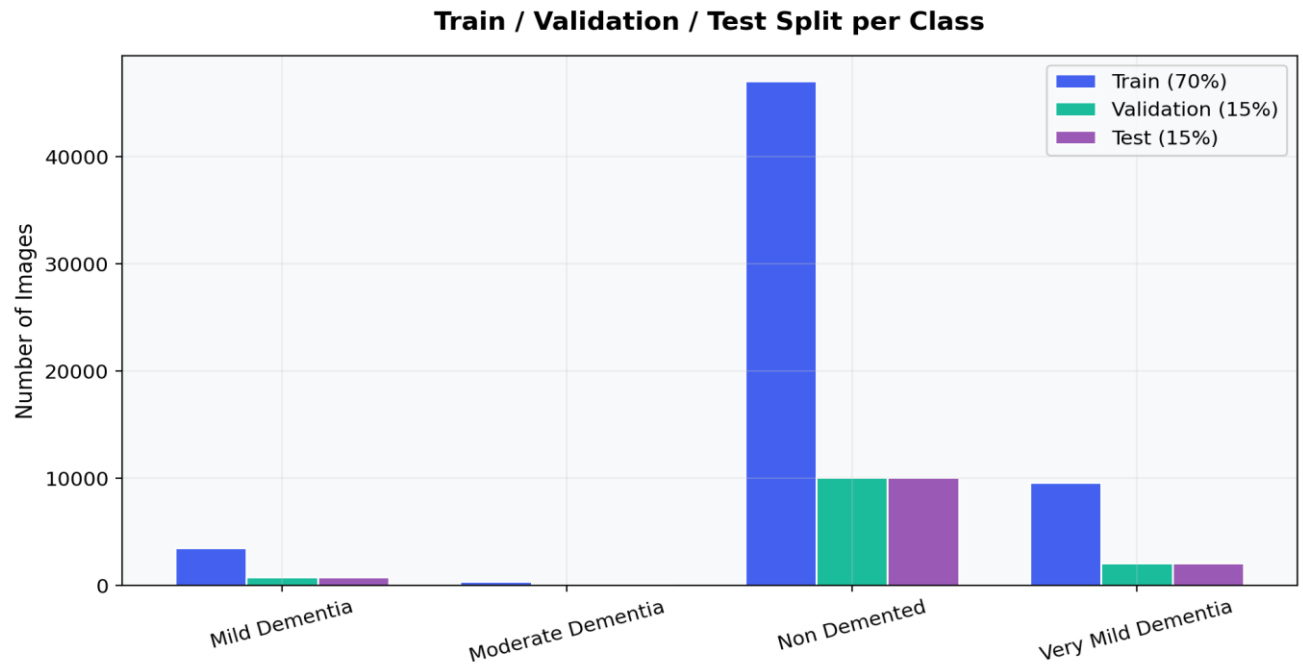
Class	Images	Share (%)	Train	Test
Mild Dementia	5,002	5.79%	3,501	751
Moderate Dementia	488	0.56%	342	73
Non Demented	67,222	77.77%	47,055	10,083

Very Mild Dementia	13,725	15.88%	9,607	2,059
TOTAL	86,437	100%	60,505	12,966



4. Data Splitting & Augmentation

The dataset was split using stratified sampling to maintain class proportions: 70% for training (60,505 images), 15% for validation (12,966), and 15% for testing (12,966). Real-time data augmentation was applied during training to improve generalization.



Augmentation Parameters

Parameter	Experiment 1	Experiment 2 (Fine-Tuned)
Rotation	20 degrees	10 degrees
Zoom	20%	10%
Horizontal Flip	Yes	No (MRI-specific)
Shear	20%	10%
Normalization	1/255	1/255

Class Weights

Moderate Dementia	73	44.2
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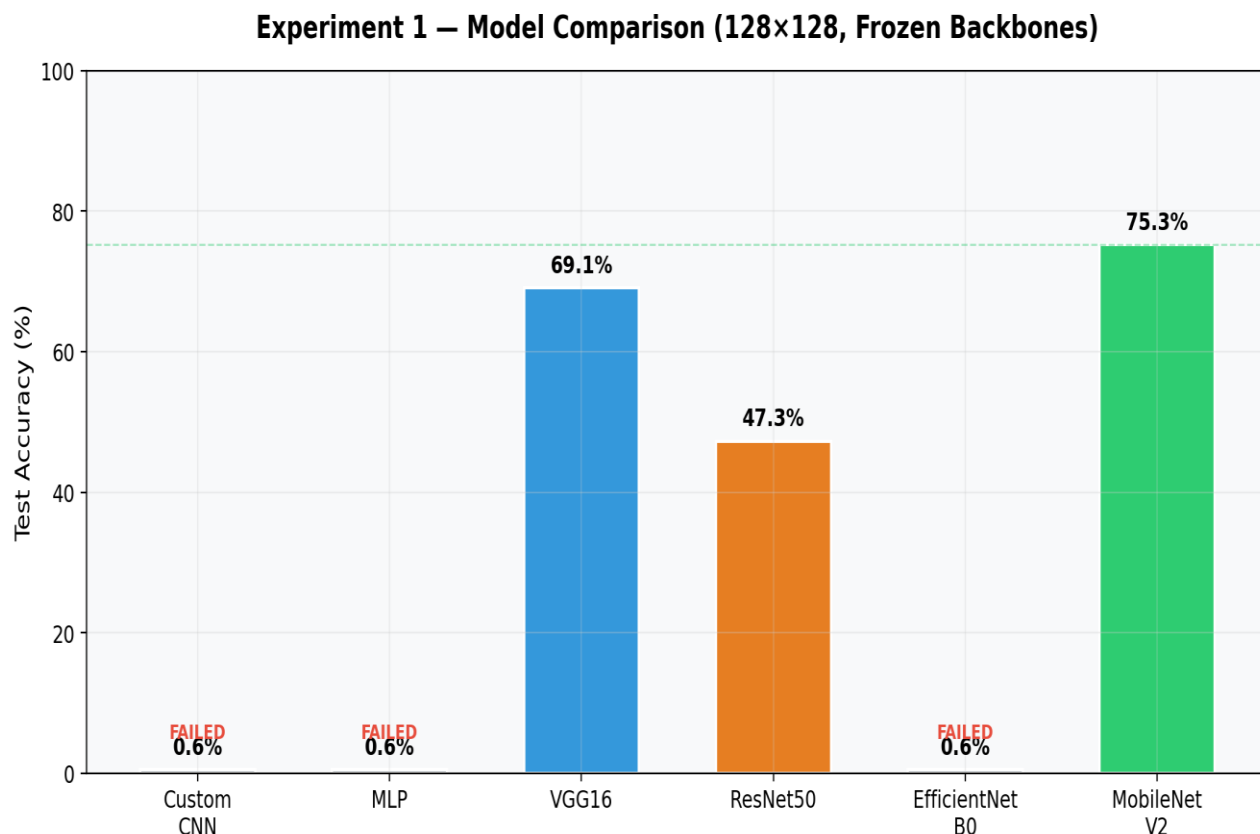
Mild Dementia	751	4.3
Very Mild Dementia	2,059	1.6
Non-Demented	10,083	0.3

5. Experiment 1 - Initial Model Comparison

Training for all six models was conducted on Kaggle notebooks using smaller 128×128 images, with all pretrained layers frozen so that only the classification head was trained. Data augmentation was more aggressive, including horizontal flipping, a rotation range of $\pm 20^\circ$, and a zoom range of 20%. No EarlyStopping or learning rate scheduling was applied. The models were trained using the Adam optimizer (learning rate = 1e-4) for 20 epochs. Under these settings, only transfer learning models with sufficiently strong feature extraction capabilities achieved meaningful accuracy.

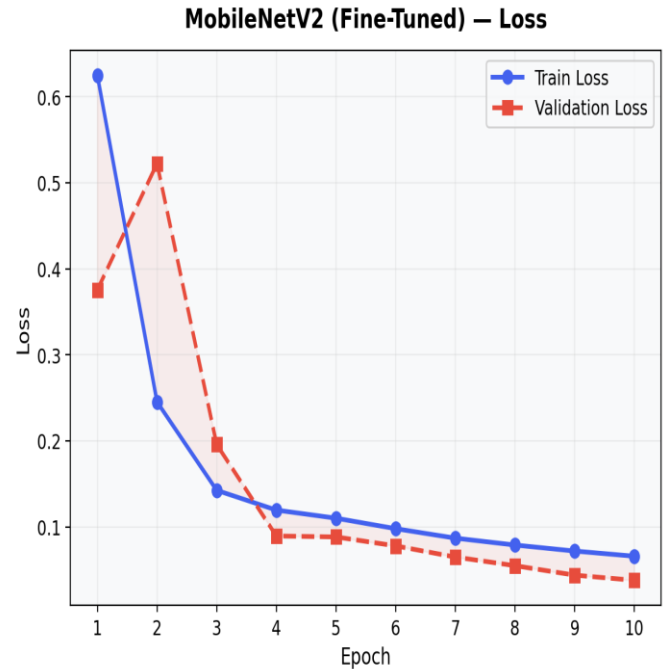
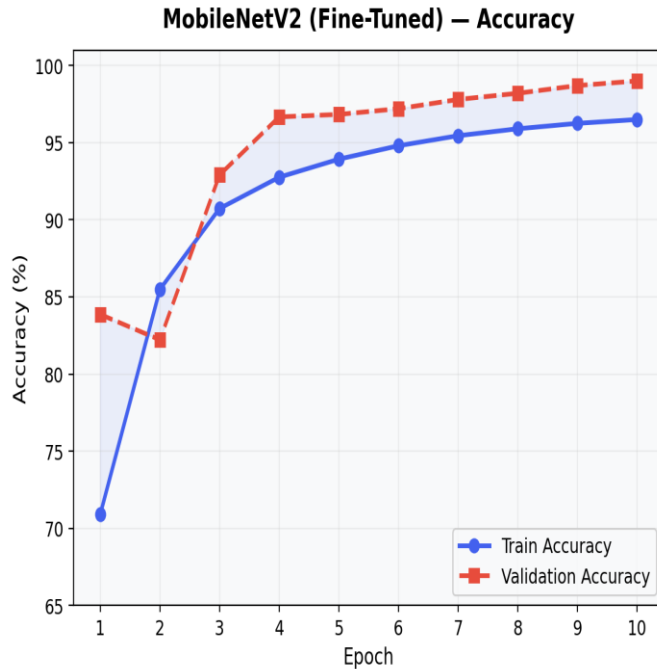
Model	Test Accuracy	Weighted F1	Macro F1	Status
MobileNetV2	75.27%	0.78	0.63	Best
VGG16	69.10%	0.73	0.44	Moderate
ResNet50	47.29%	0.56	0.26	Poor
Custom CNN	0.56%	0.00	0.00	Failed
MLP	0.56%	0.00	0.00	Failed

EfficientNetB0	0.56%	0.00	0.00	Failed
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6. Experiment 2 - Fine-Tuned MobileNetV2 (224x224)

Based on Experiment 1 results, MobileNetV2 was selected for fine-tuning with higher resolution input (224x224), partial backbone unfreezing (last 30 layers trainable), lower learning rate (1e-4), and MRI-specific augmentation (no horizontal flip). This achieved 99.47% test accuracy.



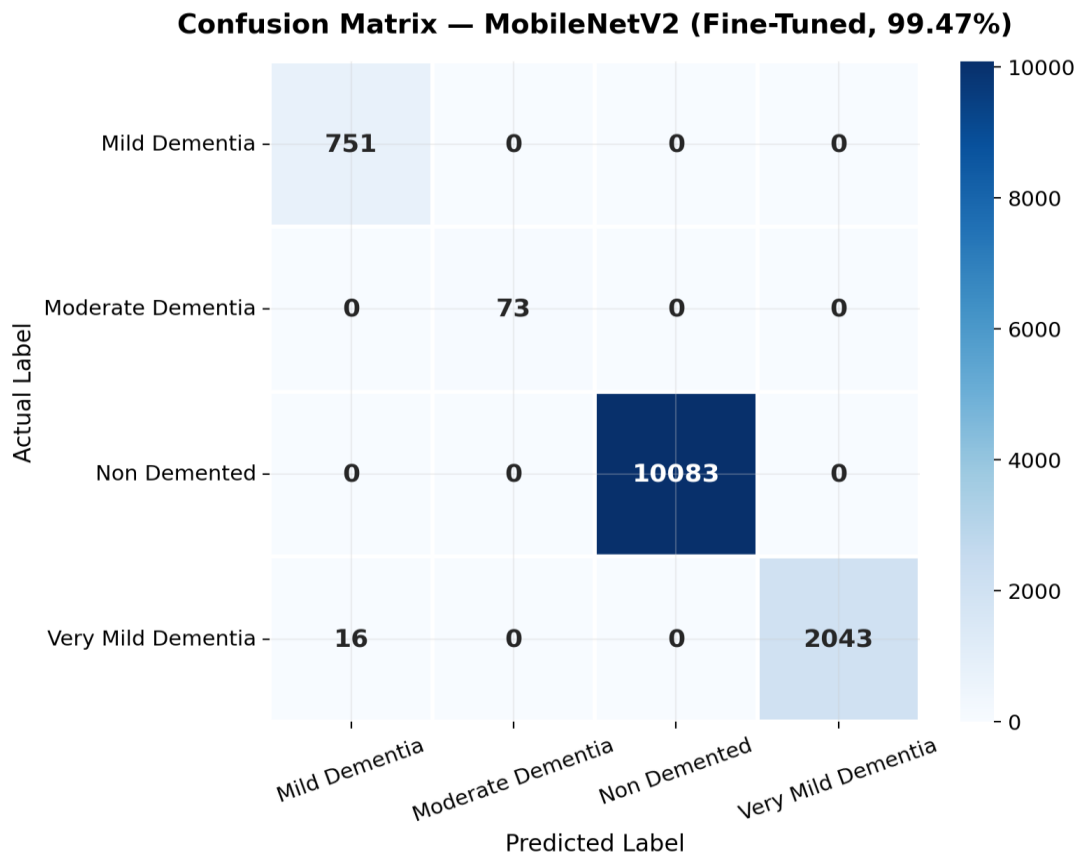
7. Performance Metrics

Classification Report - MobileNetV2 (Fine-Tuned)

Overall Test Accuracy: 99.47% on 12,966 test images.

Class	Precision	Recall	F1-Score	Support
Mild Dementia	0.98	1.00	0.99	751
Moderate Dementia	1.00	1.00	1.00	73
Non Demented	1.00	1.00	1.00	10,083
Very Mild Dementia	0.99	0.98	0.99	2,059
Weighted Average	0.99	0.99	0.99	12,966

Confusion Matrix



8. Technologies Used

Category	Technology	Purpose
Language	Python 3.12	Core programming language
Deep Learning	TensorFlow 2.19 / Keras	Model building and training
Pre-trained	MobileNetV2 (ImageNet)	Transfer learning backbone
Image Processing	OpenCV, PIL	Loading and resizing MRI scans
Data Science	NumPy, Pandas	Array operations, dataframes
Visualization	Matplotlib, Seaborn	Plots, confusion matrices
ML Utilities	scikit-learn	Splitting, metrics, class weights

Web Framework	Flask	Backend web application
Database	SQLite3	User authentication, prediction logs
Security	Werkzeug	Password hashing (PBKDF2-SHA256)
Frontend	HTML5, CSS3, JS	UI templates and interactivity
Training GPU	Google Colab (T4)	GPU-accelerated training
Dataset Source	Kaggle (kagglehub)	Automated data download

9. Web Application

A complete Flask web application provides real-time Alzheimer's detection with user authentication, MRI image upload, model selection, confidence scoring, and detailed clinical context including probable causes and treatment suggestions.

Key Features

- User Registration & Login with secure PBKDF2 password hashing.
- MRI Image Upload supporting JPG/PNG formats (max 16MB).
- Real-time Prediction with per-class probability distribution.
- Clinical Context with detailed probable causes and treatment suggestions.
- Prediction History stored in SQLite database.

Application Routes

Route	Description
/home	Home page with project statistics
/about	Disease information, dataset, technology stack

/register	New user registration
/login	User authentication
/predict	MRI upload and prediction (requires login)
/methodology	ML pipeline and training documentation

Database Schema

Table	Field	Type	Description
users	id	INTEGER PK	Auto-increment primary key
users	full_name	TEXT	User full name
users	email	TEXT UNIQUE	User email address
users	password_hash	TEXT	PBKDF2-SHA256 hashed password
predictions	id	INTEGER PK	Auto-increment primary key
predictions	user_id	INTEGER FK	References users.id
predictions	image_filename	TEXT	Uploaded MRI filename
predictions	model_used	TEXT	Model architecture used
predictions	predicted_class	TEXT	Predicted Alzheimer stage
predictions	confidence	REAL	Prediction confidence (%)

10. Conclusion

Key Findings

- Transfer learning with fine-tuning vastly outperforms training from scratch on medical imaging tasks.
- MobileNetV2 with partial backbone unfreezing achieved 99.47% test accuracy with only 2.6M parameters.
- Input resolution matters: upgrading from 128x128 to 224x224 improved accuracy by +24.20 percentage.
- Models trained from scratch (CNN, MLP) completely failed to converge on this imbalanced dataset.
- The fine-tuned model achieved perfect 1.00 F1-score on the rarest class (Moderate Dementia, only 73 test samples).
- Class weighting effectively addressed the 138:1 imbalance ratio between majority and minority classes.

Limitations

- The model is trained on a single dataset (OASIS) and should be validated externally before clinical use.
- MRI preprocessing (skull stripping, normalization) was not applied, which could further improve performance.

Future Work

- Implement Grad-CAM visualization to show which brain regions influence predictions.
- Add support for 3D volumetric MRI analysis (currently using 2D slices).
- Validate on external datasets (ADNI, MIRIAD) for clinical robustness.

11. References

- [1] OASIS Dataset: <https://www.oasis-brains.org/>
- [2] Kaggle Dataset:
<https://www.kaggle.com/datasets/ninadaithal/imagesoasis>
- [3] MobileNetV2: Sandler et al., 'MobileNetV2: Inverted Residuals and Linear Bottlenecks', CVPR 2018.
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- [7] TensorFlow: <https://www.tensorflow.org/>
- [8] Flask: <https://flask.palletsprojects.com/>